



Otto Lilienthal

German Otto Lilienthal (1848–96) was the most important pioneer of early aviation. From a young age he and his brother Gustav—his lifelong collaborator—studied the flight of birds. After the failure of his early attempts at manned flight with strapped-on wings, Lilienthal turned his hand to designing gliders. He was a qualified engineer, and his lucrative design of a small engine enabled him to pursue his passion for flight in earnest. In 1884 he built a conical hill outside Berlin from which he could launch his gliders into the wind. He constructed 18 different models of hang gliders, mostly made from stripped willow rods and tautly stretched, strong cotton fabric. These were steered by the adjustment of the rider's body to alter the center of gravity, which required significant strength.

CONTROLLED FLIGHT

While others—notably Sir George Cayley—had experimented with gliders, Lilienthal's systematic approach elevated the pursuit of flight to a new level. His designs were the first to enable sustained and replicable flights, and he was able to glide distances of up to nearly 750 ft (230 m). Lilienthal could skillfully control his craft, but it had a tendency to pitch down, partly because it was attached at the shoulder. In 1896 Lilienthal's glider stalled and nosedived, leaving him with a broken spine. His last words to his brother were “sacrifices must be made.”

This representation of a glider test flight of 1891 shows a controlled descent from Lilienthal's artificially constructed 45-ft (15-m) hill.